

Fast and Faithful Edge Bundling for Large and Complex Networks

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ABSTRACT

Edge bundling reduces the visual complexity of drawings of dense graphs by clustering “compatible” edges. However, existing edge bundling methods often have high computational complexity, leading to scalability issues. This paper presents a new framework for fast edge bundling and faithfulness metrics for large and complex graphs using *spectral sparsification*, which sparsifies a graph G into a subgraph G' with $O(n \log n)$ edges, preserving the spectrum of G . We first present a general framework *FEB* (Fast Edge Bundling) utilizing spectral sparsification to improve the efficiency of existing bundling methods while maintaining a similar quality of bundling. We then present *FBQ* (Fast Bundling Quality) framework for *proxy bundle faithfulness* metrics, to measure how *FEB* faithfully preserves the ground truth structure in the original edge bundling, with two variants, *FBQ_{JS}* (utilizing Jaccard Similarity) and *FBQ_{SQ}* (utilizing sampling quality metrics). Extensive experiments using various real-world networks demonstrate the *efficiency* of the *FEB* framework, with 61% runtime improvement over the original edge bundling methods without sparsification, while maintaining a similar quality by *FBQ* quality metrics and visual comparison.

Index Terms: Edge bundling, Spectral sparsification, Faithfulness metrics

1 INTRODUCTION

Edge bundling reduces the visual complexity in drawings of dense graphs and improves *readability* by clustering compatible edges together into “bundles” [16]. Edges are bundled together if they are “compatible”, mostly based on the geometry of edges in earlier edge bundling methods such as GBEB (Geometry-Based Edge Bundling) [3] and FDEB (Force-Directed Edge Bundling) [7]. However, bundling edges solely based on geometric properties may often introduce distortion by bundling structurally “unrelated” edges, resulting in misleading, ambiguous drawings and failing to represent the ground truth structure of the graph accurately.

Recent *structure-based* bundling methods incorporate structural properties of the graph to avoid ambiguity, improving the faithfulness of the edge bundling methods. Examples include Confluent drawings [5], TGI-EB (Topology-Geometry-Importance Edge Bundling) [23] based on graph topology and centrality, Clustered Edge Routing [1] using well-separated pair decomposition, and EPB (Edge-Path Bundling) [33] grouping edges based on paths in the graph to avoid non-existent connections. The latest SEB (Spectral Edge Bundling) method uses spectral compatibility based on the effective resistance values of graphs [13].

Note that edge bundling methods are often computationally intensive, with high runtime complexity, especially for structure-based methods. For example, FDEB runs in $O(m^2n)$ time and EPB

runs in $O(m^2 \log n)$ time, where m is the number of edges and n is the number of vertices of a graph G . Existing fast bundling methods, such as FFTEB (Fast Fourier Transform Edge Bundling) [15], whose runtime depends on the size of the bitmap image, rather than the size of the graph, still retain readability issues. The latest SEPB (Spanner EPB) [34], which runs faster than EPB in practice, still has $O(m^2 \log n)$ worst-case runtime. Therefore, there is a strong demand for fast edge bundling methods to address the scalability issues for bundling drawings of large and complex graphs with a few thousand vertices and $O(n^2)$ edges.

In this paper, we present new edge bundling approaches incorporating *SS* (*Spectral Sparsification*) for improving both the effectiveness and efficiency of edge bundling methods. Spectral sparsification [28] sparsifies the edges of a dense graph G with $O(n^2)$ edges to a subgraph G' with $O(n \log n)$ edges, based on the *effective resistance* values of edges, computed as the voltage drop along edges when modeling the graph as an electrical network [29]. Note that *SS* preserves the spectrum of G , which is closely related to important structural properties, such as connectivity, clustering, and commute distance [32, 2].

Recent works have demonstrated both the effectiveness and efficiency of spectral sparsification (*SS*) for sampling and drawing large and complex graphs, outperforming popular random sampling methods on various sampling quality metrics [6, 20, 10, 9]. Moreover, *SS* has been successfully integrated with the two most popular graph drawing algorithms, force-directed and stress minimization methods, to improve the force (resp., stress) computation from quadratic to sublinear, while maintaining good quality drawings [18, 21]. Therefore, it is a natural approach to leverage spectral sparsification to improve the efficiency and effectiveness of edge bundling of drawings of large and complex graphs.

In this paper, we first present the *FEB* (*Fast Edge Bundling*) framework, which utilizes spectral sparsification [30] to improve the runtime efficiency of existing edge bundling methods while maintaining a similar quality to the original bundling.

We then present the *FBQ* (*Fast Bundling Quality*) metrics for measuring *proxy bundling faithfulness*, i.e., how the *FEB* framework faithfully preserves the ground truth structure in the original edge bundling, based on Jaccard Similarity and sampling quality metrics. Our main contributions can be summarized as follows:

1. We present a fast edge bundling framework, *FEB*, based on *spectral sparsification*, computing edge bundling on a smaller subgraph to reduce runtime while maintaining a similar quality of bundling. Extensive experiments demonstrate the efficiency and effectiveness of *FEB*, running 61% faster than the original edge bundling methods without sparsification while maintaining similar quality using visual comparison.
2. We present the *FBQ* framework to measure how faithfully the *FEB* framework preserves the ground truth structures of the original bundling without sparsification. We present *FBQ_{JS}*, using *JS* (*Jaccard similarity*), and *FBQ_{SQ}*, based on sampling quality metrics. Experiments show the effectiveness of *FBQ* metrics for accurately measuring the faithfulness of the *FEB* framework, consistent with visual comparison.

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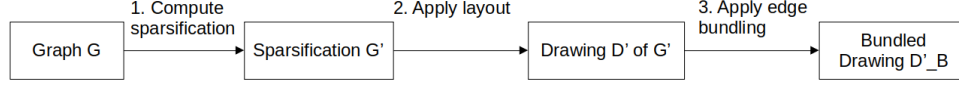


Figure 1: FEB Framework.

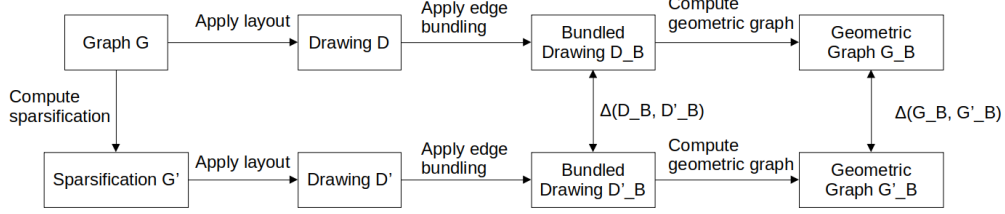


Figure 2: FBQ Faithfulness Metrics framework.

2 FEB (FAST EDGE BUNDLING) FRAMEWORK

Graph sampling is a popular approach to reducing the runtime for drawing large and complex graphs G , i.e., drawing a smaller sample G' of G is much faster than drawing G . In particular, spectral sparsification (SS) has demonstrated effectiveness for sampling, outperforming random sampling on sampling quality metrics [6, 20, 10, 9] and efficiency for drawing (i.e., drawing graphs fast preserving similar quality) [18, 21]. We therefore leverage SS for efficient and effective edge bundling on the drawings of large and complex graphs.

Specifically, we present the *FEB* framework, which computes edge bundling on the drawing D' of a sparsification $G' \subset G$, instead of the drawing D of the original graph G . Figure 1 shows the FEB framework consisting of the following steps:

1. Compute a sparsification $G' = (V, E')$ of $G = (V, E)$.
2. Compute a drawing D' of G' .
3. Compute a bundled drawing D'_B of D' .

In Step 2, a drawing is computed by applying a graph drawing algorithm (or using a given layout for geographic networks).

We present a specific instance of the FEB framework leveraging SS. As SS reduces the number of edges of dense graphs with $O(n^2)$ edges to $O(n \log n)$ edges, and successfully improves the runtime of drawings of large complex graphs [18, 21], we expect that SS can significantly improve the runtime of edge bundling algorithms (i.e., D'_B of G' can be computed much faster than D_B of G).

Moreover, as SS computes a sparsification G' , preserving important structural properties, such as connectivity, clustering, and commute distance [30], and outperforms random sampling methods [20, 10, 9], we expect a bundled drawing D'_B of G' using SS to maintain high similarity to the bundled drawing D_B of G .

3 FBQ (FAST BUNDLING QUALITY) METRICS

We present the *FBQ* (Fast Bundling Quality) metrics for evaluating the effectiveness of the FEB framework. We measure *proxy bundling faithfulness*, i.e., how faithfully the bundled drawing D'_B computed by FEB preserves the structures of the original bundled drawing D_B without sparsification, by extending the proxy faithfulness metric for evaluating drawings of graph samples [24].

Figure 2 shows the FBQ framework, where we measure the effectiveness of FEB (i.e., the similarity between D_B and D'_B) by computing the similarity between G_B and G'_B , geometric graphs computed from D_B and D'_B . The FBQ framework is computed as:

1. Compute a sparsification G' of G .
2. Compute drawings D of G and D' of G' .

3. Compute bundled drawings D_B of D and D'_B of D' .

4. Construct geometric graphs G_B of D_B and G'_B of D'_B .

5. Compare the similarity between D_B and D'_B (i.e., $\Delta(D_B, D'_B)$) using the similarity between G_B and G'_B (i.e., $\Delta(G_B, G'_B)$).

In Step 2, a drawing is computed by applying a graph drawing algorithm (or using a given layout for geographic networks).

In Step 4, we construct a *geometric graph* from the bundled drawing as follows [33]. Consider a bundled drawing D_B , where edges $e_1 = (s, t)$ and $e_2 = (u, v)$ are bundled together, with vertices s and u on the same end of the bundle and t and v are on the other end of the bundle. Once the edges are bundled, not only would the endpoints of each edge e_1 and e_2 appear as neighbors, but also the pairs (s, v) and (t, u) . The geometric graph G_B thus contains four edges $(s, t), (u, v), (s, v), (t, u)$, including the edges between these pairs of vertices “implied” as neighbors by bundling.

For Step 5, we present two variants of the *FBQ* metric, based on how to measure the similarity between G_B and G'_B (i.e., $\Delta(G_B, G'_B)$). The first variant *FBQ_{JS}* is based on the *Jaccard Similarity* (JS) [12], computed as follows:

$$FBQ_{JS}(G_B, G'_B) = \frac{1}{|V|} \sum_{v \in V} \frac{|N_B(v) \cap N'_B(v)|}{|N_B(v) \cup N'_B(v)|}$$

where $N_B(v)$ (resp., $N'_B(v)$) is the neighbors of v in G_B (resp., G'_B).

The second variant *FBQ_{SQ}* is based on the *sampling quality metrics*, which computes the similarity between a sample graph and the original graph using important structural graph properties [11]. Specifically, the similarity between G_B and G'_B is computed using *Kolmogorov-Smirnov (KS) distance*, which measures the similarity between two cumulative distribution functions (CDFs) as follows:

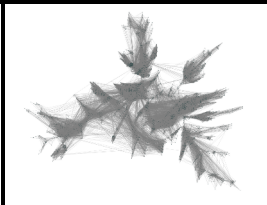
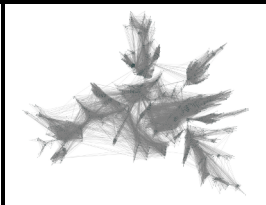
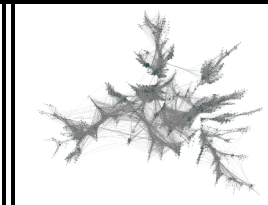
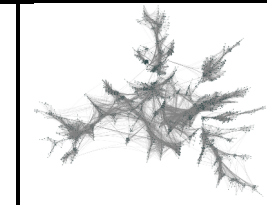
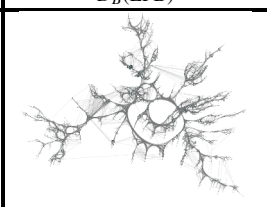
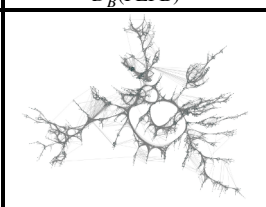
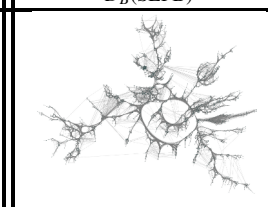
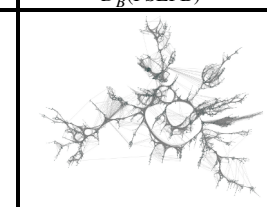
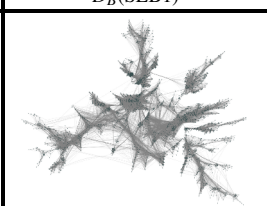
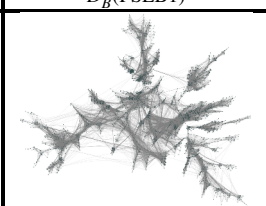
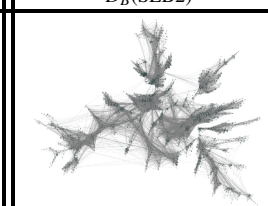
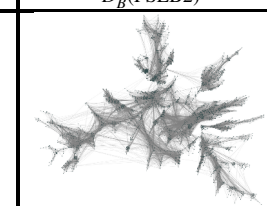
$$FBQ_{SQ} = KS(SQ(G_B), SQ(G'_B))$$

where $SQ(G_B)$ (resp., $SQ(G'_B)$) is the CDF of a selected graph property computed on G_B (resp., G'_B).

More specifically, we define *FBQ* based on the graph properties, popular sampling quality metrics *SQ*, as follows:

- *FBQ_{SQ(DG)}*: based on the *Degree Correlation Associativity (DEG)*, which measures the likelihood that vertices connect to other vertices with similar degrees [22].
- *FBQ_{SQ(CC)}*: using *Clustering Coefficient (CC)*, which measures the tendency of vertices to be clustered together [27].

Table 1: Visual comparison between FEB and the original edge bundling methods for the scale-free Facebook graph. For all FEB methods, the bundled drawing D'_B by FEB is visually almost the same as the corresponding original bundled drawing D_B .

D	D'	$D_B(\text{FDEB})$	$D'_B(\text{FFDEB})$
			
$D_B(\text{EPB})$	$D'_B(\text{FEPB})$	$D_B(\text{SEPB})$	$D'_B(\text{FSEPB})$
			
$D_B(\text{SEB1})$	$D'_B(\text{FSEB1})$	$D_B(\text{SEB2})$	$D'_B(\text{FSEB2})$
			

4 FEB EXPERIMENTS

We now present extensive experiments for evaluating the efficiency and effectiveness of the FEB framework (i.e., significantly faster than the original bundling methods without sparsification, with a similar quality). *SS* reduces the number of edges of the graph to be bundled, thereby improving the runtime efficiency, while preserving important structural properties of the graph, enabling the bundled drawing to represent the original edge bundling faithfully.

We use various benchmark data sets from previous edge bundling and graph drawing studies: (i) real-world *geographic* graphs with fixed positions assigned to each vertex [33]; (ii) real-world *scale-free* graphs with globally sparse, locally dense clusters with short diameters [4, 26]; (iii) *GION* graphs, biochemical networks with globally sparse (path-like), locally dense structures with long diameters [17]; and (iv) *black-hole* graphs with globally sparse (mesh- or cycle-like) and locally dense “blobs” structures [6]. Table 2 shows the details.

	type	$ V $	$ E $	den	dia
airline	geographic	235	1297	8.940	4
migration	geographic	1702	6487	5.714	12
airtraffic	geographic	1533	16480	10.759	6
yeastppi	scale-free	2224	6609	2.972	11
facebook	scale-free	4039	88234	21.846	8
lastfm	scale-free	7624	27806	3.647	15
6_gion	GION	1785	20459	11.462	41
7_gion	GION	3010	41757	13.873	77
G443	black-hole	285	2009	7.049	12
Cycle896	black-hole	1031	22638	21.957	35
Cycle907	black-hole	823	14995	18.220	42

Table 2: Data sets including the number of vertices ($|V|$), the number of edges ($|E|$), density (den), and diameter (dia).

To draw graphs in (ii)-(iv) without given geometry, we use the *Backbone* layout [25], designed to untangle “hairballs” in drawings of large and complex graphs, shown to produce highly faithful drawings by shape-based and cluster-faithfulness metrics [8, 19].

We denote FEB using an original bundling method *X* as *FX* (i.e., FEB with SEB is denoted as FSEB). For each bundling method, we compute bundled drawings D_B (resp., D'_B) from a drawing D (resp., D') of the original graph G (resp., the sparsification G').

Specifically, we consider five edge bundling methods, including the original Force-Directed Edge Bundling (FDEB) [7] method as well as the state-of-the-art methods Edge-Path Bundling (EPB) [33] and Spanner-based Edge-Path Bundling (SEPB) [34], which have been shown to generally obtain lower ambiguity than other bundling methods, such as Confluent drawings [5], CUBu [31], and Winding Roads [14], especially on higher orders of the ambiguity metric. Moreover, we also include the latest SEB (Spectral Edge Bundling) method [13] with two variations, SEB1 and SEB2, which improve the FDEB method by introducing *spectral compatibility* using the effective resistance values.

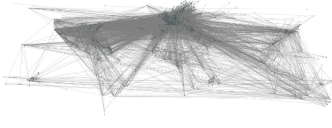
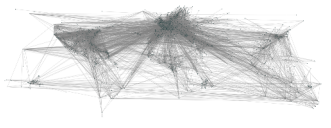
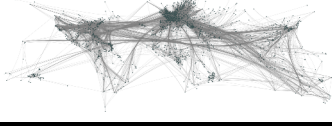
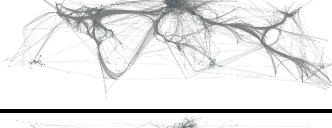
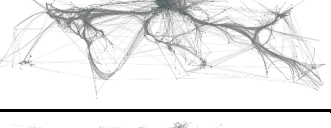
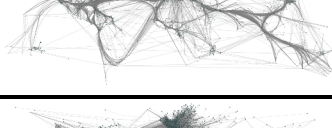
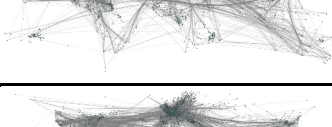
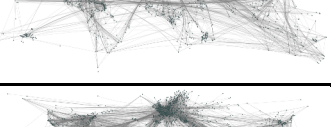
4.1 Runtime Comparison

Figure 3(a) shows the runtime improvement of FEB methods, computed as $\frac{t(X)-t(FX)}{t(X)}$, where $t(X)$ is the runtime by the original bundling method *X* and $t(FX)$ is the runtime by FEB using *X*.

On average, FEB obtains 61% runtime improvement over the original bundling methods. Note that FEB with the FDEB family (i.e., FFDEB, FSEB1, FSEB2) obtains 68% runtime improvement on average, where FSEB2 obtains the highest improvement at 70%.

FEB with the EPB family (i.e., FEPB, FSEPB) obtains 53% runtime improvement on average, due to SEPB having 35% improvement since it is designed as a fast variation of EPB.

Table 3: Visual comparison for the Air Traffic graph, with original edge bundling methods on the left column and FEB on the right column. For all FEB methods, the bundled drawing D'_B by FEB is visually almost the same as the corresponding original bundled drawing D_B .

	D		D'
			
	D_B		D'_B
FDEB		FFDEB	
EPB		FEPB	
SEPB		FSEPB	
SEB1		FSEB1	
SEB2		FSEB2	

4.2 Quality Metrics Comparison

Figure 3(b) demonstrates the high similarity between the bundled drawings computed by FEB (D'_B) and the original bundled drawings (D_B), with FBQ_{JS} on average 68%. Overall, FEB with SEB methods performs better than FEB with EPB methods on most graphs: 20% better on scale-free and black-hole data sets on average.

Figures 3(c)-(d) show even higher similarity between the bundled drawings by FEB (D'_B) and the original bundled drawings (D_B), with $FBQ_{SQ(DG)}$ and $FBQ_{SQ(CC)}$. On both metrics, FEB obtains 0.26 (i.e., 74% similarity to the original bundled drawings) on average, showing that the bundled drawings by FEB faithfully display the structure of the underlying graph to a similar extent as the original bundled drawings. FSEB methods obtain slightly better $FBQ_{SQ(DG)}$ over FEB with the EPB family (EPB, SEPB), which obtain slightly better $FBQ_{SQ(CC)}$.

4.3 Visual Comparison

Visual comparisons also show the consistent results, i.e., FEB computes visually highly similar bundled drawings to the original bundled drawings. For example, Table 1 shows visual comparisons between FEB and the original edge bundling methods on a scale-free graph, Facebook. For all FEB methods, the bundled drawing D'_B by FEB is visually almost the same as the corresponding original bundled drawing D_B .

Table 3 shows visual comparisons between FEB and the original edge bundling methods on a geographic graph, Air Traffic. Overall, the bundled drawings D'_B by FEB are visually almost the same as

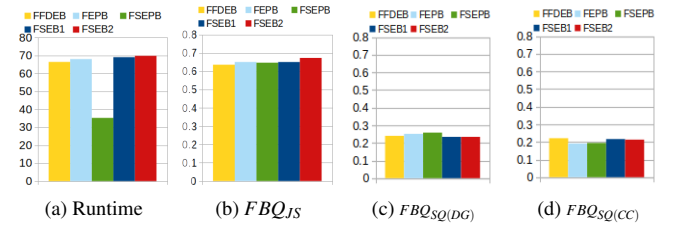


Figure 3: (a) 61% runtime improvement by FEB methods over original bundling methods; (b) FBQ_{JS} (higher=better) by FEB methods, on average 0.68, and (c-d) FBQ_{SQ} (lower=better) by FEB methods, on average 0.26, showing high similarity of D_B and D'_B .

the corresponding original bundled drawing D_B , maintaining highly similar structures to D_B , with large bundles mostly preserved.

4.4 Summary

We present a new FEB framework for fast edge bundling and new FBQ metrics to measure the faithfulness of the FEB framework. Extensive experiments demonstrate the efficiency (i.e., 61% runtime improvement) and effectiveness (i.e., highly similar to the original edge bundling, 74% by FBQ_{SQ}) of the FEB framework. Visual comparisons show consistent results with the FBQ metrics.

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